

1.

Fisher's linear discriminant is

$$\hat{\mathbf{w}}^* = \arg \max_{\hat{\mathbf{w}}} J(\hat{\mathbf{w}}) = \arg \max_{\hat{\mathbf{w}}} \frac{\hat{\mathbf{w}}^T \mathbf{S}_b \hat{\mathbf{w}}}{\hat{\mathbf{w}}^T \mathbf{S}_w \hat{\mathbf{w}}}$$

where  $\mathbf{S}_b = (\mathbf{m}_1 - \mathbf{m}_2)(\mathbf{m}_1 - \mathbf{m}_2)^T$  and  $\mathbf{S}_w = \sum_j \sum_{\alpha} (\mathbf{x}_{\alpha} - \mathbf{m}_j)(\mathbf{x}_{\alpha} - \mathbf{m}_j)^T$

(a)

By writing  $\frac{dJ(\hat{\mathbf{w}})}{d\hat{\mathbf{w}}} = 0$ , show that

$$\mathbf{S}_w^{-1} \mathbf{S}_b \hat{\mathbf{w}} = J(\hat{\mathbf{w}}) \hat{\mathbf{w}}.$$

(b)

Explain why  $\hat{\mathbf{w}}^*$  is the eigenvector for which  $J(\hat{\mathbf{w}})$  is the maximum eigenvalue of  $\mathbf{S}_w^{-1} \mathbf{S}_b$ .

(c)

Explain why  $\mathbf{S}_b \hat{\mathbf{w}}$  is always in the direction of  $\mathbf{m}_1 - \mathbf{m}_2$  and thus show that

$$\hat{\mathbf{w}}^* = \text{const.} \cdot \mathbf{S}_w^{-1} (\mathbf{m}_1 - \mathbf{m}_2)$$

2.

Another way to optimize Fisher's linear discriminant (suggested by Barry Fridling):

(a)

Show that for any two real vectors  $\mathbf{x}$  and  $\mathbf{y}$

$$(\mathbf{x}^T \mathbf{y})^2 \leq (\mathbf{x}^T \mathbf{x})(\mathbf{y}^T \mathbf{y}), \quad (\text{Cacuhy-Schwarz})$$

(b)

Show that if the  $\lambda_k$  are positive,

$$\left( \sum_{k=1}^N x_k y_k \right)^2 \leq \left( \sum_{k=1}^N \lambda_k x_k^2 \right) \left( \sum_{k=1}^N y_k^2 / \lambda_k \right).$$

(c)

Thus, show that for  $\mathbf{A}$  positive definite

$$(\mathbf{x}^T \mathbf{y})^2 \leq (\mathbf{x}^T \mathbf{A} \mathbf{x})(\mathbf{y}^T \mathbf{A}^{-1} \mathbf{y}).$$

(d)

By letting  $\mathbf{A} = \mathbf{S}_w$  in the expression above, and writing

$$J(\hat{\mathbf{w}}) = \frac{|\hat{\mathbf{w}}^T (\mathbf{m}_1 - \mathbf{m}_2)|^2}{\hat{\mathbf{w}}^T \mathbf{S}_w \hat{\mathbf{w}}}$$

show again that

$$\hat{\mathbf{w}}^* = \text{const.} \cdot \mathbf{S}_w^{-1}(\mathbf{m}_1 - \mathbf{m}_2).$$

### 3.

Using the Optdigits dataset from the UCI repository, implement PCA. Reconstruct the digit images and calculate the error  $E(n) = \sum_{j=1}^m \|\hat{\mathbf{x}}_{j,n} - \mathbf{x}_j\|^2$  for various values of  $n$ , the number of principal components included in the reconstruction of each image. Plot  $E(n)$  versus  $n$ .